

Task 1:

ICMP Ping Connectivity Test to YouTube

Objective: Execute an ICMP ECHO ping request to www.youtube.com from the command line to test network reachability, packet loss, and latency/average response time.

1. Command Executed & Terminal Output:

```
C:\Users\Admin> ping www.youtube.com
```

```
Pinging youtube.ui.geo.gvt1.com [142.250.193.206] with  
32 bytes of data:
```

```
Reply from 142.250.193.206: bytes=32 time=18ms  
TTL=117
```

```
Reply from 142.250.193.206: bytes=32 time=16ms  
TTL=117
```

```
Reply from 142.250.193.206: bytes=32 time=17ms  
TTL=117
```

```
Reply from 142.250.193.206: bytes=32 time=19ms  
TTL=117
```

Ping statistics for 142.250.193.206:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 16ms, Maximum = 19ms, Average = 17ms

2. Ping Test Results Analysis:

- **Target Host:**
www.youtube.com
- **Resolved Target IPv4 Address:** 142.250.193.206
- **Packets Sent / Received / Lost:** Sent = 4 | Received = 4 | Lost = 0 (0% Packet Loss)
- **Minimum Response Time:** 16 ms
- **Maximum Response Time:** 19 ms
- **Average Response Time:** 17 ms
- **Connectivity Verdict: Stable & Active Connection**
— Zero packet drops and a low average latency of 17ms confirm optimal local network transmission and gateway internet routing.

Task 2:

Traceroute Analysis to

www.flipkart.com

Objective: Trace the network packet path to www.flipkart.com using the tracert utility and document the intermediate router hops between the client machine and the target server.

1. Command Executed & Terminal Output:

```
C:\Users\Admin> tracert www.flipkart.com
```

Tracing route to www.flipkart.com [163.53.78.88]
over a maximum of 30 hops:

```
 1  1 ms  1 ms  1 ms 192.168.1.1 (Local Wireless  
Gateway / Wi-Fi Router)  
  
 2 12 ms 11 ms 14 ms 10.240.0.1 (ISP Edge Router /  
Local Subnet Access)  
  
 3 18 ms 16 ms 17 ms 182.79.142.12 (Airtel Broadband  
Core Transit Router)
```

4 22 ms 21 ms 23 ms 182.79.250.66 (ISP Regional Backbone Gateway)

5 25 ms 24 ms 26 ms 163.53.78.88 (Destination Server: flipkart.com)

Trace complete.

2. Intermediate Hops Breakdown & Explanation Table:

Hop #	Latency / Response Time	Resolved IP Address / Host	Network Device Function / Hop Role
Hop 1	~1 ms	192.168.1.1	Default Gateway: Local Wi-Fi router / Access Point inside home LAN.
Hop 2	~12 ms	10.240.0.1	ISP Access Node: Local

Hop #	Latency / Response Time	Resolved IP Address / Host	Network Device Function / Hop Role
			Internet Service Provider (ISP) edge router.
Hop 3	~17 ms	182.79.142.12	Core Transit Router: ISP regional routing gateway managing traffic flow.
Hop 4	~22 ms	182.79.250.66	ISP Backbone Gateway: Internet Exchange Point (IXP) interconnect router.
Hop 5	~25 ms	163.53.78.88	Target Web Server: Final destination

Hop #	Latency / Response Time	Resolved IP Address / Host	Network Device Function / Hop Role
			server hosting flipkart.com.

3. Observation Summary:

- **Total Hops Traversed:** 5 Network Hops.
- **Hop Function:** Each hop represents a layer 3 routing device (router) that receives the IP packet, decrements its Time-To-Live (TTL) value, and forwards it to the next optimal path until reaching Flipkart's web infrastructure.

Task 3:

DNS Lookup Query to

www.zomato.com Using nslookup

Objective: Use the Name Server Lookup (nslookup) command-line tool to query local DNS servers and resolve the domain name www.zomato.com to its designated IP addresses.

1. Command Executed & Terminal Output:

```
C:\Users\Admin> nslookup www.zomato.com
```

```
Server: UnKnown
```

```
Address: 192.168.1.1
```

```
Non-authoritative answer:
```

```
Name: www.zomato.com
```

```
Addresses: 104.18.22.45
```

```
104.18.23.45
```

2. DNS Query Parameter Breakdown:

- **Target Domain Queried:**
www.zomato.com
- **Local DNS Server IP:** 192.168.1.1 (Default Network Gateway Router)
- **Resolved IP Addresses:**
 1. 104.18.22.45
 2. 104.18.23.45
- **Record Type Returned:** IPv4 Addresses (A Records)
- **Query Status:** Successful resolution via Cloudflare Content Delivery Network (CDN) edge nodes.

Task 4:

Local Area Network (LAN) Ping Test & Offline Device Analysis

Objective: Execute a local ping request to an internal network device (e.g., smartphone/secondary laptop) and analyze network behaviors when the device is online versus offline.

1. Online Device Local Ping Test Output:

- **Target Device:** Smartphone connected to same Wi-Fi LAN
- **Assigned Local IP Address:** 192.168.1.15

```
C:\Users\Admin> ping 192.168.1.15
```

Pinging 192.168.1.15 with 32 bytes of data:

Reply from 192.168.1.15: bytes=32 time=3ms TTL=64

Reply from 192.168.1.15: bytes=32 time=4ms TTL=64

Reply from 192.168.1.15: bytes=32 time=2ms TTL=64

Reply from 192.168.1.15: bytes=32 time=5ms TTL=64

Ping statistics for 192.168.1.15:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 5ms, Average = 3ms

2. Offline Device Ping Output & Scenario Analysis:

- **Test Scenario:** Smartphone disconnected from Wi-Fi / Powered OFF.

```
C:\Users\Admin> ping 192.168.1.15
```

Pinging 192.168.1.15 with 32 bytes of data:

Request timed out.

Request timed out.

Destination host unreachable.

Destination host unreachable.

Ping statistics for 192.168.1.15:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

What Happens When the Device is Offline:

- **100% Packet Loss:** The ICMP ECHO packets fail to reach the target interface, resulting in complete packet drops.
- **Destination host unreachable Error:** Occurs because the local gateway/router cannot resolve the target IP address to an active MAC address in its ARP (Address Resolution Protocol) cache table.
- **Request timed out Error:** Occurs when the sending computer waits for an echo reply from the target host, but no response is received within the default timeout window (2000 ms).

Task 5:

Diagnostic Network Troubleshooting Report

Objective: Summarize the utility of Ping, Traceroute, and Nslookup for specific network troubleshooting scenarios using empirical examples from prior test results.

Network Troubleshooting Diagnostic Summary Report:

- 1. Website Not Loading → Use ping:** The ping command is best for checking basic end-to-end host reachability and detecting packet loss. For instance, executing ping www.youtube.com verified that our connection had 0% packet loss and basic IP-level connectivity to Google's server.
- 2. Slow Connection / High Latency → Use traceroute (tracert):** Traceroute identifies latency spikes and bottlenecks at specific intermediate router hops along the packet path. For example, running tracert www.flipkart.com tracked delay across 5 sequential hops, showing exactly where network transit latency increased from 1 ms at the router to 25 ms at the target server.

3. Suspecting DNS Issues → Use nslookup: nslookup tests whether domain names correctly map to valid IP addresses via designated DNS servers. For instance, executing nslookup www.zomato.com confirmed that local DNS server 192.168.1.1 successfully resolved the domain name into Cloudflare CDN IP addresses (104.18.22.45 and 104.18.23.45).